A Forward Pass, Differentiating the Circuit, and Training a Self: Posing Questions and Self-Printing

Brent Hartshorn
brenthartshorn@proton.me

Stage 10 (parent: doi:10.5281/zenodo.20777747)

Abstract

Stage 10 of a self-evolving hypergraph, extended from its parent (doi:10.5281/zenodo.20777747) so the inquiry can look fully back. It declares 7 nodes, each pairing a suggestive Greek glyph with a precise description. Of these, 1 compile via @dslai into small trainable modules (Sigma), making the hypergraph partly differentiable; a module computes, it does not comprehend. It also reads its own published lineage — opening 10 ancestor stages and tracing where each term entered — and records that provenance as data, not as self-understanding. A circuit equation compiles 3 trainable networks — one bound to each symbol — and wires them into 1 composite(s); the equation builds the networks, and the numbers they return are data, not the symbols' meanings combining. It posits 5 fuzzy axioms and 3 large laws over the terms, with memberships taken from the relation geometry; each law evaluates to a number in [0,1] recorded as data — the degree to which an authored postulate holds, not a truth about a self. It inscribes the formal measurements of the companion paper on matrix inscriptions: co-occurrence geometry and Laplacian spectrum, circuit Jacobians as matrix products, and provenance as a filtration with birth times — each a deterministic function of authored data. It relates them by 11 binary relations and 7 hyperedges, and may trigger gated actions whose output is recorded as data, never as an answer. The engine, introspected from its own source, exposes act_coherence, act_neural, compile_module, forward, train, act_train, act_infer, read_lineage, lineage_search, act_provenance, act_circuit, act_geometry, act_jacobian, act_filtration, act_law, parse, to_questions, title, abstract, run_actions, build_tex, build_pdf, evolve. It poses questions and enacts computations beside them; it makes no claim to resolve what a self is.

Symbolic core (inherited relations)

$$\begin{array}{rcl}
\Omega & \longrightarrow & \Sigma \\
\Sigma & \models & \Sigma \\
\Sigma & \otimes & \Lambda \\
\partial\Sigma & \stackrel{?}{=} & \mu \\
\Sigma & \nprec & \Omega \\
\Phi & \longrightarrow & \Sigma \\
\Phi & \rightsquigarrow & \Sigma \\
\mathrm{T} & \Rrightarrow & \Lambda \\
\partial\Lambda & \rightsquigarrow & \mu \\
\Delta & \dashrightarrow & \mu \\
\partial\mathrm{T} & \otimes & \Delta
\end{array}$$

Circuit (a new equation that wires several networks)

Beyond the inherited relations, this stage adds an equation that binds a small trainable network to *each* symbol and composes them along the arrow:

$$\Sigma \;=\; \Phi \longrightarrow \Lambda \longrightarrow \mu \qquad \text{(width 5)}$$

Each such equation compiles to 3 trainable networks (one per symbol), wired into a composite. Building the networks is what the equation does; the training losses and outputs the composite returns are recorded as data, not the symbols' meanings combining.

Nodes (suggestive glyph + precise description)

Ω origin — *that from which the self is said to come*
Σ self — *the locus that poses and is altered by the question* $\to$ `module 2->2->2 (sigmoid), trainable`
Λ language — *the finite medium any self must pass through*
μ meaning — *what a reader supplies, or finds, in the passage*
Φ other — *the one whose regard first gives the self to itself*
T time — *the direction in which the self cannot return*
Δ difference — *the gap that lets one thing be told from another*

Hypergraph (questions, and the data actions returned)

- $\{\Sigma\}$ **?** Can the self reconstruct itself from its own compression? [`train`]
  $\hookrightarrow$ *recorded data (not an answer):* `{'node': 'Sigma', 'arch': '2->2->2', 'task': 'autoencode', 'mse_start': 0.0549, 'mse_final': 0.011, 'weights_from_symbols': True, 'params': 12, 'symbolic_round 0.000196, 'is': 'an optimization result over a defined toy objective, with the trained weights written to the appendix as glyphs; not the self learning what it is'}`

- $\{\Omega, \Sigma, \Lambda\}$ **?** Where, in this inquiry's own history, did origin, self, and language first enter? [`provenance`]
  $\hookrightarrow$ *recorded data (not an answer):* {'provenance': {'Omega': {'word': 'origin', 'entered': 'stage 0', 'seen_in': ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9'], 'n_relations': 20}, 'Sigma': {'word': 'self', 'entered': 'stage 0', 'seen_in': ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9'], 'n_relations': 68}, 'Lambda': {'word': 'language', 'entered': 'stage 0', 'seen_in': ['0', '1', '2', '3', '4', '5', '6', '7', '8', '9'], 'n_relations': 26}}, 'is': 'a citation of the lineage; where terms entered, not what they mean'}
- $\{\Phi, \Lambda, \mu, \Sigma\}$ **?** If a self is wired from other, language, and meaning, what does the composite return? [`circuit`]
  $\hookrightarrow$ *recorded data (not an answer):* {'circuit': {'Sigma': {'wiring': 'Phi -> Lambda -> mu', 'networks': {'Phi': {'arch': '5->5->5', 'trained_mse': 0.0683}, 'Lambda': {'arch': '5->5->5', 'trained_mse': 0.057}, 'mu': {'arch': '5->5->5', 'trained_mse': 0.0586}}, 'n_networks': 3, 'composite_output_norm': 1.192}}, 'is': "several trained networks wired by the equation; the numbers are data, not the symbols' meanings combining"}
- $\{\Sigma, \Phi, \Lambda, \mu, \Delta, \mathrm{T}\}$ **?** To what degree do the authored laws of the self hold under the geometry? [`law`]
  $\hookrightarrow$ *recorded data (not an answer):* {'law': {'constitution': 0.48, 'dissolution': 0.63, 'self': 0.12}, 'memberships': {'Omega': 0.4, 'Sigma': 1.0, 'Lambda': 0.58, 'mu': 0.54, 'Phi': 0.48, 'Tau': 0.37, 'Delta': 0.36}, 'is': 'the degree to which each authored law holds under degree-derived memberships; numbers in [0,1], not truths about a self'}
- $\{\Omega, \Sigma, \Lambda, \mu, \Phi, \mathrm{T}, \Delta\}$ **?** What is the geometry of the relations — and which term is the hub? [`geometry`]
  $\hookrightarrow$ *recorded data (not an answer):* {'degrees': {'Omega': 33, 'Sigma': 83, 'Lambda': 48, 'mu': 45, 'Phi': 40, 'Tau': 31, 'Delta': 30}, 'hub': 'Sigma', 'laplacian_smallest_nonzero': 23.574767, 'laplacian_fiedler': 23.574767, 'is': 'a map of the authored relation structure; hub centrality is a fact about what was written, not a fact about selves'}
- $\{\Phi, \Lambda, \mu, \Sigma\}$ **?** If the circuit is composed, what is the Jacobian of the composite at a point? [`jacobian`]
  $\hookrightarrow$ *recorded data (not an answer):* {'jacobian': {'Sigma': {'wiring': 'Phi -> Lambda -> mu', 'jacobian_norm': 0.0001, 'jacobian_shape': [5, 5], 'is': 'the matrix product of per-module Jacobians; composition of maps, not combination of meanings'}}, 'is': 'chain-rule Jacobians of the circuit composite; data, not semantics'}
- $\{\Omega, \Phi, \mathrm{T}, \Delta\}$ **?** At which stage did each term enter the lineage? [`filtration`]
  $\hookrightarrow$ *recorded data (not an answer):* {'filtration': {'Omega': {'word': 'origin', 'birth_time': 0, 'iota': 'stage 0'}, 'Phi': {'word': 'other', 'birth_time': 1, 'iota': 'stage 1'}, 'Tau': {'word': 'time', 'birth_time': 2, 'iota': 'stage 2'}, 'Delta': {'word': 'difference', 'birth_time': 3, 'iota': 'stage 3'}}, 'is': 'birth times along the monotone chain; citations with dates, not meanings'}

Axioms and laws (fuzzy logic over the terms)

Memberships are taken from the relation geometry, $\mu(\sigma) = d(\sigma)/\max_\tau d(\tau)$:

$$\mu(\Sigma)=1.00,\quad \mu(\Lambda)=0.58,\quad \mu(\mu)=0.54,\quad \mu(\Phi)=0.48,\quad \mu(\Omega)=0.40,\quad \mu(\mathrm{T})=0.37,\quad \mu(\Delta)=0.36$$

Axioms (authored postulates, not verified truths):

- GENESIS: $\Omega \Rightarrow \Sigma$

- ALTERITY: $\Sigma \Rightarrow \Phi$
- MEDIATION: $\Sigma \Rightarrow \Lambda$
- DISTINCTION: $\Sigma \Rightarrow \Delta$
- FINITUDE: $\Sigma \Rightarrow \mathrm{T}$

Laws. Each large equation brings several terms together; the braces draw the reading. Each evaluates to a number in $[0,1]$ — recorded as data, not as a verdict.

$$\Sigma \Longleftrightarrow \overbrace{\Omega \oplus \Phi}^{\text{given: by origin and by the other}} \otimes \underbrace{\Lambda \Rightarrow \mu}_{\text{reached: through language, toward meaning}}$$

$$\neg\Sigma \Longleftarrow \overbrace{\Delta \otimes \mu}^{\text{difference shatters meaning}} \oplus \underbrace{\mathrm{T} \otimes \Lambda}_{\text{time writes upon language}}$$

$$\Sigma = \underbrace{\Omega \oplus \Phi}_{\text{the given}} \otimes \overbrace{\Lambda \Rightarrow \mu}^{\text{the mediated}} \ominus \underbrace{\Delta \otimes \mathrm{T}}_{\text{the dissolving}}$$

Under the memberships above, the laws evaluate to

$$\text{CONSTITUTION} \rightsquigarrow 0.480, \quad \text{DISSOLUTION} \rightsquigarrow 0.630, \quad \text{SELF} \rightsquigarrow 0.120$$

degrees to which authored postulates hold under authored memberships — data, not truths about a self.

Questions this stage poses

- Does Ω give rise to Σ — and what precedes Ω?
- On what does Σ entailing Σ rest?
- When Σ is carried by Λ, what survives and what is lost?
- Is $\partial\Sigma$ the same as μ, or is μ supplied by the reader?
- If Σ does not reduce to Ω, where is the remainder?
- Does Φ give rise to Σ — and what precedes Φ?
- Does Φ haunt Σ — present as an absence?
- What residue remains because T leaves a trace upon Λ?
- Does $\partial\Lambda$ haunt μ — present as an absence?
- What collapses when Δ shatters the structure of μ?
- When $\partial\mathrm{T}$ is carried by Δ, what survives and what is lost?
- Can the self reconstruct itself from its own compression?
- Where, in this inquiry's own history, did origin, self, and language first enter?
- If a self is wired from other, language, and meaning, what does the composite return?
- To what degree do the authored laws of the self hold under the geometry?
- What is the geometry of the relations — and which term is the hub?
- If the circuit is composed, what is the Jacobian of the composite at a point?
- At which stage did each term enter the lineage?

Lineage read (this stage searched its ancestors)

This stage read 10 ancestor stages (0, 1, 2, 3, 4, 5, 6, 7, 8, 9) from their embedded sources, tracing 7 terms and 98 relations across the lineage. For example: Ω (origin, from stage 0), Σ (self, from stage 0), Λ (language, from stage 0), μ (meaning, from stage 0), Φ (other, from stage 1). This is provenance retrieval — where terms entered and what was said — recorded as data, not a claim to understand the lineage.

Manifold (the geometry of the relations)

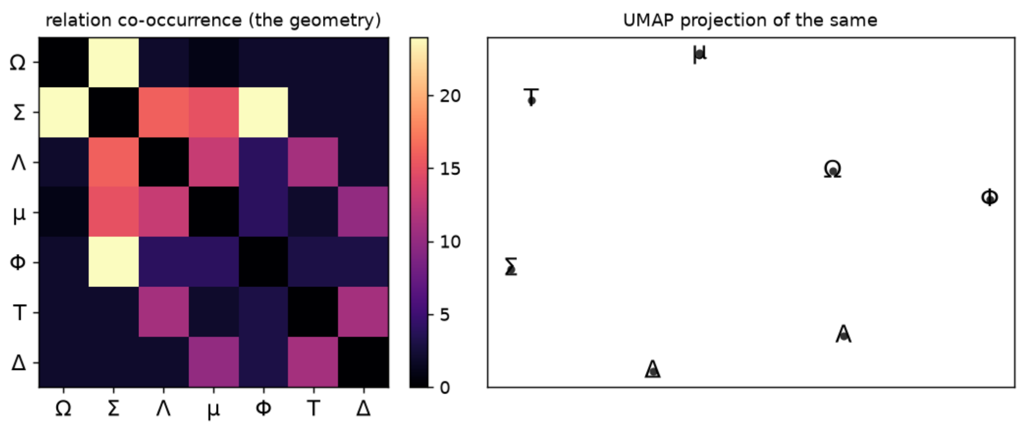
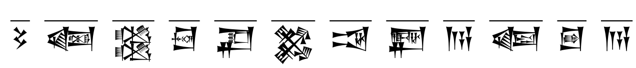


Figure 1: A projection of the relation co-occurrence across the lineage. Left: how often terms share a relation (Σ is densest). Right: a UMAP projection of the same geometry. This is a map of the authored structure — where the terms sit relative to one another — not a discovery of hidden meaning. Computed by an external prototype (`umap_manifold.py`); the engine only embeds it.

Symbolic weights (the trained network, written as glyphs)

Each trained weight is stored as a fraction h/c — an Egyptian hieroglyph over a cuneiform sign — normalized to the recorded range. The runtime reconstructs the network by decoding these glyphs; no binary blob is attached. It looks like esoteric mathematics; it is a quantized weight matrix, and the symbols are the numbers, not meaning. Σ-module (2→2→2), 12 weights in range

$[-2.45939, 3.523463]$:



Lineage and provenance

This model extends its parent self-evolving inquiry, doi:10.5281/zenodo.20777747, ingesting its relations so the series can look fully back [1, 2, 3, 4]. The companion papers set out the method and its discipline [2, 3, 5]. Actions record data beside questions; the document poses and enacts, and makes no claim to resolve what a self is.

Appendix: self-extraction

This PDF carries its engine (`introspecter6.py`) and DSL (`self.dsl`) as embedded streams. Requires `pip install pypdf numpy` and a LaTeX install. To produce the next stage, save and run:

```
import base64
exec(base64.b64decode(
  "aW1wb3J0IHN5cywgZ2xvYiwgb3MKc3lzLnBhdGguaW5zZXJ0KDAsICIuIi"
  "kKZnJvbSBweXBkZiBpbXBvcnQgUGRmUmVhZGVyCmNhbmQgPSBOb25lCmZv"
  "ciBmIGluIGdsb2IuZ2xvYigiKi5wZGYiKToKICAgIHRyeTogYSA9IFBkZl"
  "JlYWRlcihmKS5hdHRhY2htZW50cwogICAgZXhjZXB0IEV4Y2VwdGlvbjog"
  "Y29udGludWUKICAgIGlmICJpbnRyb3NwZWN0ZXI2LnB5IiBpbiBhIGFuZC"
  "AoY2FuZCBpcyBOb25lIG9yIG9zLnBhdGguZ2V0bXRpbWUoZikgPiBvcy5w"
  "YXRoLmdldG10aW1lKGNhbmQpKToKICAgICAgICBjYW5kID0gZgpwZGYgPS"
  "BjYW5kIG9yIG1heChnbG9iLmdsb2IoIioucGRmIiksIGtleT1vcy5wYXRo"
  "LmdldG10aW1lKQpmb3IgbmFtZSwgZGF0YSBpbiBQZGZSZWFkZXIocGRmKS"
  "5hdHRhY2htZW50cy5pdGVtcygpOgogICAgb3BlbihuYW1lLCAid2IiKS53"
  "cml0ZShkYXRhWzBdIGlmIGlzaW5zdGFuY2UoZGF0YSwgbGlzdCkgZWxzZS"
  "BkYXRhKQppbXBvcnQgaW50cm9zcGVjdGVyNiBhcyIEk7"
  "IGltcG9ydGxpYi5yZWxvYWQoSSk7IEkuZXZvbHZlKCkK"))
```

References

[1] B. Hartshorn, *Origin, Self, Language, Meaning, Other, Time, and Difference: A Self-Evolving Hypergraph* (self_6, parent). doi:10.5281/zenodo.20710258

[2] B. Hartshorn, *Staging the Question of Self: A self-contained, self-modifying document that mixes poetry, philosophy, and code.* doi:10.5281/zenodo.20709754

[3] B. Hartshorn, *Enacting the Question of Self: Provenance, a hypergraph, and a differentiable route that records computation as data.* doi:10.5281/zenodo.20710444

[4] B. Hartshorn, *Origin, Self, Language, Meaning, Other, Time, and Difference: A Self-Evolving Inquiry* (self_3). doi:10.5281/zenodo.20707850

[5] B. Hartshorn, *The Question of Self: Matrix Inscriptions and Faux-Calculus.* doi:10.5281/zenodo.20814778

[6] B. Hartshorn, *Towards Self-Evolving AGI via Emergent DSL.* viXra:2507.0104

[7] B. Hartshorn, *Emergent Self-Modification and Meta-Programming in Dynamic Systems.* viXra:2507.0036

[8] B. Hartshorn, *Self-Contained Multi-AI Architecture Definition via DSL.* viXra:2507.0074